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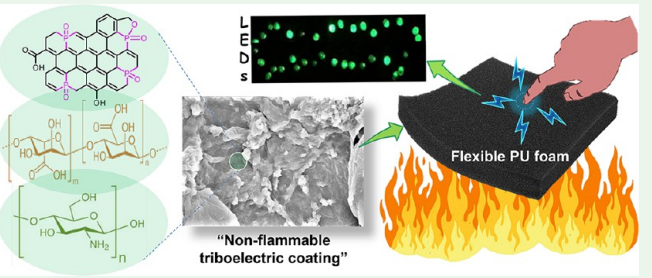
Polyurethane Foams Coated with Phosphorus-Doped Mesoporous Carbon for Flame-Retardant Triboelectric Nanogenerators

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ABSTRACT: A phosphorus-doped mesoporous carbon (PMC)-filled coating was deposited on polyurethane foam (PUF) via a layer-by-layer assembly method. First, PMC was prepared from bio-sourced saccharose and phytic acid using mesoporous silica KIT-6 as a hard template. The coated PUF was then prepared by alternate dipping into a chitosan (CH) solution and dipping into an alginate-stabilized PMC (AL-PMC) aqueous dispersion. A few layers (three bilayers) of the CH/AL-PMC coating allowed the PUF to self-extinguish when exposed to a butane flame (≈1400 °C). Additionally, this coating layer enabled the foam to pass the UL-94 rating. Cone calorimetry revealed that this coating reduced the peak heat release rate, rate of smoke release, total smoke release, peak CO_2 production rate, and peak CO production rate by 56, 48, 29, 35, and 35%, respectively. Using this foam, a flame retardant foam-based triboelectric nanogenerator (FRF-TENG) was fabricated. The FRF-TENG exhibited excellent energy harvesting performance, offering 158 V and 2.26 μA cm^{-2} open-circuit voltage and short-circuit current density, respectively. Furthermore, the FRF-TENG can be attached on a chair and serve as a self-powered sensor for the detection of back movement and sit-stand motion. This study may provide a promising potential for the design and fabrication of multifunctional smart foams.



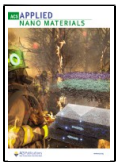
KEYWORDS: *ordered mesoporous carbon, layer-by-layer assembly, polyurethane foam, flame retardancy, triboelectric nanogenerator*

1. INTRODUCTION

Polyurethane foam (PUF) is light weight and comfortable and possesses good resilience and insulating properties. Thus, it is ubiquitous in seat cushions, mattresses, packaging, vehicle seats, and heat, electricity, or noise insulators.[1−6] Additionally, recent studies have extended the functionality and usefulness of PUFs by developing smart foams (or electrical foams).[7−11] In this regard, foam-based triboelectric nanogenerators (TENGs) capable of converting all forms of mechanical energy into electricity have been started to develop. Owing to the inherent flexibility, compressibility, convenient fabrication process, and easy integration into cushions and seating chairs, foam-based TENGs hold a promising future in several smart applications, including sitting posture, child or disabled people safety, and traffic safety monitoring.[11] However, the vast majority of foams are victims of fire owing to PUF's high flammability,[12] which greatly limits their use in the above-mentioned application fields. When PUF is ignited, it generates a flammable melt pool that swiftly transfers fire to neighboring flammable materials,[1] and 90% of its mass is paralyzed into harmful smoke and gases.[12] Hence, incorporating flame retardancy into PUF is crucial to secure life and property. To date, various flame retardants (FRs) have been developed to modify the flammability of PUF either by adding FR fillers during fabrication or back-finishing using different surface coating techniques (i.e., layer-by-layer (LbL) assembly).[6] Compared to the former treatment method, the latter is facile, versatile, and eco-friendly[13] because it imposes minimal impact on PUF's intrinsic properties,[14] conformally coats its complex 3D structure,[6,15] and allows for precise control of coating thickness and composition.[16] Above all, the LbL coating enables the incorporation of multiple FR materials, including nanoparticles, macromolecules, and biomolecules, into a single FR system to achieve synergy.[17,18]

Carbon materials such as graphene oxide (GO),[19−22] carbon nanotubes (CNTs),[23−25] and carbon nanofibers (CNFs)[26] are prospective candidates for the fabrication of FR multilayers by LbL on PUF. Among these carbon materials, FRs filled with GO or CNTs display excellent performance in enhancing the fire safety of PUF. For instance, GO has been combined with both alginate (AL) and chitosan (CH)[19] or CH alone[22] to prevent foam collapse and melt dripping and substantially reduce the peak heat release rate (pkHRR). Other studies have

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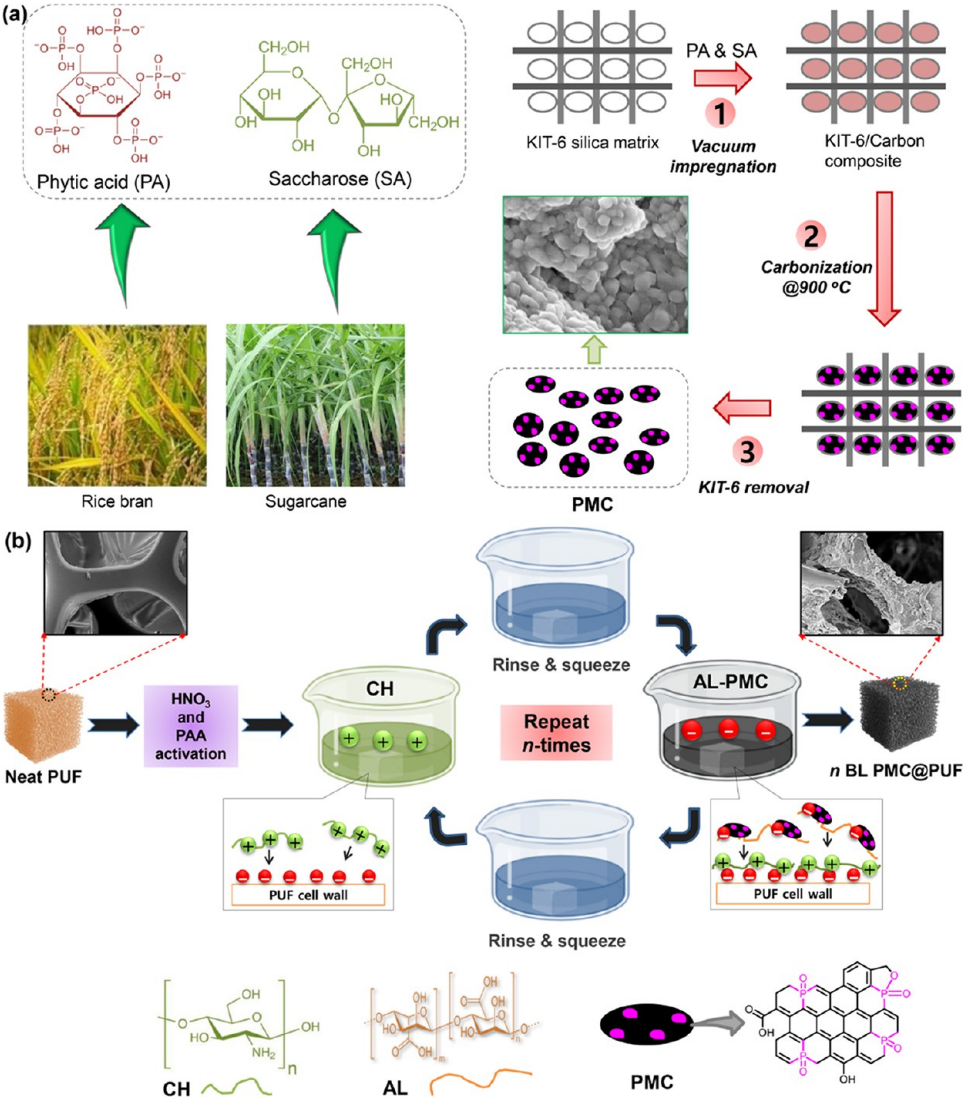


Figure 1. (a) Synthesis procedure for PMC and (b) schematic of the LbL assembly approach for deposition of the CH/AL-PMC coating on PUF. PUF was pre-activated by sequentially dipping in HNO_3 (0.1 N) and polyacrylic acid (PAA) (1 wt %) prior to the LbL process.

demonstrated that a CNT-filled multilayer of CH, AL, and montmorillonite (MMT) deposited on PUF[23,25] substantially reduces both the pkHRR (by 65−69%) and total smoke release rate (TSR, by 80%). However, these approaches merely focused on achieving high FR performance and neglected the following limitations, which are vital for the advancement and scalability of the method. (i) Most of the coatings require a high number of deposition steps (up to 30 layers) to achieve the desired properties, making them, expensive, time-consuming, and tedious.[19,20,23,24] (ii) Some approaches use a high concentration of phosphate-based salts as an additive to reduce the number of steps, but adding salt further complicates the process and imposes additional cost.[16] (iii) One-step deposition-based methods achieve excellent fire retardancy by sacrificing mechanical properties of the foam due to the much higher FR add-on (80% to the PUF's weight).[27] (iv) The commonly used GO- and CNT-based FRs are non-renewable as they originate from petroleum resources[28] and expensive due to complicated synthesis and functionalization steps.[29−31]

To address these limitations, in this study, we examine the potential of phosphorus-doped mesoporous carbon (PMC) as a surface modification material to improve both FR and triboelectric properties of PUF. First, PMC was prepared using saccharose (SA) as a carbon source along with phytic acid (PA), which served as a precursor for both phosphorous and carbon and ordered mesoporous silica (KIT-6) as a hard template (Figure 1a). Next, a PMC-filled coating comprising a negatively charged, AL-stabilized PMC and CH as the positively charged counterpart was constructed on the PUF cells via the LbL process (Figure 1b). Porous carbons are part of a new carbon family that has gained widespread attention owing to their large specific surface area, good thermal and mechanical stability, and low density.[32] Porous carbons are prepared by a simple and convenient synthesis method using cheap, ubiquitous, and sustainable biomass precursors. Most importantly, porous carbon materials can achieve excellent flame retardancy owing to the ease of introducing heteroatoms such as phosphorous (P) on the surface of the carbon framework and/or inside the porous channels during the synthesis process.[33−35] These materials have been demonstrated excellent performance in the fields of catalysis, gas storage, and electrochemical energy storage and conversion systems.[29] Here, for the first time, renewable porous carbon is used as a building block of LbL assembly for the surface

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engineering of PUF. The synthesized porous carbon material and coating film were thoroughly characterized by several analytical techniques. The FR performances of different coating layers on PUF were studied using an open flame test and cone calorimetry. Only three bilayers (BLs) of CH/AL-PMC coated PUF prevented ignition when exposed to a butane flame (≈1400 °C) while also significantly reducing the key FR metrics, such as pkHRR, RSR, TSR, peak CO_2 production rate, and peak CO production rate by 56, 48, 29, 35, and 35%, respectively. Additionally, the 3 BL CH/AL-PMC coating significantly enhanced the TENG performance of the foam. The TENG could be attached on a chair, generating outputs of 7 and 10 V upon back movement and sit-stand motion, respectively, and the output was sufficient to concomitantly glow 21 LEDs. Compared to the coating techniques mentioned above, our coating strategy has some distinctive advantages: (i) the coating is multifunctional, meaning that the coating not only improves the flame retardancy of PUF but also imparts a triboelectric property, which allows the use of PUF in advanced and smart applications. (ii) The FR agent we explored is cheap and biomass-derived carbon instead of the expensive and non-renewable conventionally investigated GO- and CNT-based FRs. (iii) The coating process requires a few layers and does not involve salt fortification, which is beneficial for scaling up.

2. EXPERIMENTAL SECTION

2.1. Materials. Saccharose (SA, ≥99%), tetraethyl orthosilicate (TEOS, 98%), pluronic P123, triblock poly(ethylene oxide)−poly(propylene oxide)−poly(ethylene oxide), ($EO_{20}PO_{70}EO_{20}$, molecular weight 5800 g mol^{-1}), hydrofluoric acid (48% HF), phytic acid (PA, 50 wt % in water), chitosan (CH, 190,000−310,000 g mol^{-1}, 75−85% degree of deacetylation), sodium alginate (AL, 120,000−190,000 g mol^{-1}), polyacrylic acid (PAA, 100,000 g mol^{-1}), and sodium hydroxide (NaOH, ≥98.0%) were purchased from Sigma-Aldrich (MO, USA). Hydrochloric acid (HCl, 35.0%), nitric acid (HNO_3, 69.0%), and *n*-butanol (BuOH, 99.5%) were procured from Daejung (Gyeonggi-do, Korea). All chemicals were used as received. Flexible polyurethane foam (PUF) (tradename: RJBB, 0.027 g cm^{-3}) was purchased from Kyunggi Sponge (Gyeonggi-do, Korea). Ultrapure deionized (DI) water (Milli-Q Millipore, 18.2 MΩ cm conductivity at 25 °C, pH = 6.0) was utilized in this study.

2.2. Synthesis of Mesoporous KIT-6. KIT-6 was synthesized according to the procedure described elsewhere.[33] Briefly, surfactant P123 (6.0 g) was added into a mixture of HCl (11.8 g) and DI water (217.0 g) and stirred at 35 °C until the P123 was completely dissolved. Subsequently, BuOH (6.0 g) and TEOS (12.9 g) were added, and the mixture was stirred for 24 h at 35 °C. Then, the resulting mixture was hydrothermally treated at 100 °C for 24 h in a firmly closed polypropylene bottle without stirring. Next, the solution was filtered, washed with HCl/ethanol mixture, and dried at 80 °C overnight. Finally, a surfactant-free KIT-6 was obtained after calcining at 550 °C for 5 h (at a temperature ramp rate of 3 °C min^{-1}) in an air atmosphere.

2.3. Synthesis of Phosphorous-Doped Mesoporous Carbon (PMC). The PMC was prepared using SA as a carbon source, PA as a phosphorous precursor, and KIT-6 as a hard template described by Panja et al.[33] (Figure 1a). SA (6 g) and PA (4 g) were added to a small amount of DI water and vigorously stirred until a homogeneous mixture was obtained. The mixture was then vacuum-infiltrated into the mesopores of 4 g of KIT-6 at room temperature. The resulting KIT-6/carbon composite was dried at 80 °C, ground to a fine composite powder, and subsequently carbonized in a tubular furnace at 900 °C for 5 h (at a temperature ramp rate of 3 °C min^{-1}) under a N_2 atmosphere. Afterward, the KIT-6 template was removed by etching in HF (1.0 M) to obtain silica-free PMC. Undoped mesoporous carbon (MC) was also prepared similarly but without PA addition.

2.4. Layer-by-Layer Deposition. A 0.1 N HNO_3 and 1 wt % PAA aqueous solution were used for the surface treatment of PUF. Cationic CH solution (0.25 wt %) was prepared by adding CH powder to DI water, and the pH was adjusted to 6.0 with concentrated NaOH (5.0 N) and HCl (5.0 N). The AL-stabilized PMC anionic suspension (AL-PMC) containing 0.3 wt % AL and 0.5 wt % PMC was prepared by dispersing 3 g of AL and 5 g of PMC in 1 L of DI water and exfoliating by a tip sonicator (HD-1070, Sonoplus) set at 60% amplitude for 3 h. The pH was adjusted to 8.0, and the mixture was rolled for 24 h before use. AL-stabilized MC suspension was also prepared by the same process.

The LbL coating process of PUF is depicted in Figure 1b. The PUF was cut into the desired dimensions, washed thoroughly with DI water, and dried in an oven at 70 °C before the LbL coating. First, the dried PUF was dipped sequentially in 0.1 N HNO_3 and 1.0 wt % PAA solutions for 30 s in each solution to increase adhesion by introducing a net negative charge on the foam surface.[13] Thereafter, the foam was soaked in CH solution followed by compressing five times to effectively infiltrate the CH solution into the foam pores. After dipping for 2 min, the PUF was squeezed using a tube wringer and thoroughly rinsed with DI water by compressing several times to remove loosely attached CH. Next, the foam was exposed to the AL-PMC anionic solution using a similar process to deposit the first bilayer (1 BL). This process was repeated with a soaking of 1 min in each solution until the desired number of BLs was deposited. The coated samples were dried overnight in an oven at 60 °C and stored in a dry bag before testing. Foam coated with MC multilayers was also prepared in the same fashion to act as a control.

2.5. Characterization. The synthesized porous carbons were characterized by powder X-ray diffraction (XRD), N_2 adsorption−desorption isotherm, X-ray photoelectron spectroscopy (XPS), field emission SEM (FE-SEM), and high-resolution transmission electron microscopy (HR-TEM). HR-TEM images were recorded using a JEM-2100F (JEOL) at 200 kV with a 2 k × 2 k CCD camera (Gatan, Ultra Scan 1000). XRD was observed using a Bruker AXS D8 focus X-ray diffractometer (Cu-Kα radiation, step size of 0.03° at 4° min^{-1} scan speed) from 10 to 90°. N_2 adsorption−desorption isotherms were acquired using a Belsorp-mini II (Bel Japan, Inc.) at −196 °C in the relative pressure range (p/p^o) of 0.01−0.1. Specific surface areas were determined using the Brunauer−Emmett−Teller (BET) theory and pore size distribution using the Barrett−Joyner−Halenda (BJH) method. To determine the bond nature and surface elemental compositions of PMC and MC, XPS analysis was carried out using a Theta Probe AR-XPS system (Thermo Fisher Scientific, UK) (spot size ≈ 400 μm) equipped with a monochromatic Al Kα X-ray source ($h\nu$ = 1486.6 eV) at 12 kV/72 W. The spectra were processed using CasaXPS Version 2.3.19PR1.0, fitted with Lorentzian asymmetric line shapes LA (α, β) after Shirley background subtraction. PMC and MC were further scrutinized under a scanning electron microscope equipped with energy-dispersive X-ray spectroscopy (SEM−EDX, Hitachi SU-70, Japan) operating at 10.0 kV. The surface charge of the carbon materials was measured using a Zetasizer Nano ZS (Malvern, South Korea). The phosphorous content in the samples was determined using the ammonium molybdate method.[36] A reference standard curve was constructed using 1.0−10.0 mg L^{-1} anhydrous KH_2PO_4, and the absorbance at 420 nm was recorded. Elemental compositions (wt % C, H, O) were quantified using an elemental analyzer (Flash 2000 Thermo Fisher Scientific, USA). The growth of the LbL assembly was monitored by UV−vis, quartz crystal microbalance (QCM), and attenuated total reflectance−Fourier transform infrared (ATR-FTIR). Light absorption spectra were recorded by UV−vis (DH-2000-BAL, Oceans Optics). The coating was performed on plasma (PDC 32G-2, Harrick Plasma, NY, USA)-etched (10 min) glass slides. The deposited weight per layer was determined with a QCM (QCM200, SRS, Sunnyvale, California, USA). QCM crystals were first plasma-treated for 5 min. Then, the LbL deposition on the treated QCM crystals was conducted in a way like an assembly on PUF. ATR-FTIR was recorded on a Nicolet iS5

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FTIR spectrophotometer (Thermo Fisher Scientific, Korea) using 16 scans and 4 cm^{-1} resolution in the region of 500−4000 cm^{-1}. The morphologies of neat and coated PUFs were examined using FE-SEM. Samples were Au sputter-coated (4 nm thickness) before imaging. TGA of the foam sample in an air atmosphere (50 mL min^{-1}) was examined using a thermogravimetric analyzer (Mettler Toledo, TGA2, Korea). The samples (≈4 mg) were heated from 30 to 800 °C at a heating rate of 10 °C min^{-1}. To qualitatively evaluate the flammability of the coated foams relative to the pristine foam, foams (5.0 × 5.0 × 2.5 cm^3) were exposed to a butane hand torch for 10 s. The UL-94 horizontal burning test of the foams (130 × 13 × 5 mm^3) was conducted on a UL94 flame chamber (FTT, UK) according to ASTM D3801. Char residues after the torch test were analyzed by FE-SEM. Cone calorimetry was conducted as per the ISO 5660-1 standard using foam dimensions of 10.2 × 10.2 × 2.5 cm^3 and a heat flux of 35 kW m^{-2} (Fire Testing Technology, West Sussex, UK). The sampling interval was 3 s. Compressive strength was performed on a universal testing machine (LLOYD LF Plus) by compressing to a deformation rate of 60% for 500 cycles using a crosshead speed of 20 mm min^{-1}.

2.6. FRF-TENG Design and Characterizations. The FRF-TENG was simply fabricated by attaching an adhesive aluminum (Al) foil electrode onto the FR foam (5.0 × 5.0 × 2.5 cm^3), which served as a +ve triboelectric layer. Then, copper wire was attached to the Al electrode for electrical output measurements. A polytetrafluoroethylene (PTFE) film, sticked to a human finger via double-sided adhesive tape, was used as a −ve triboelectric contacting layer. The contact force and frequency were fixed at nearly ≈20 N and ≈6 Hz, respectively. The open-circuit voltage of the FRF-TENG was recorded using an oscilloscope (TDS 2012B, Tektronix), while a preamplifier (SR570, Stanford Research Systems) was used to measure the short-circuit current.

3. RESULTS AND DISCUSSION

3.1. Characterizations of PMC. The XRD patterns of PMC and MC are presented in Figure S1. Both samples showed two low-intensity, broad diffraction peaks centered at 2θ = 21 and 43°, which are assigned to the (0 0 2) and (1 0 0) crystal planes, respectively. This indicates that a porous carbon framework with an amorphous graphitic structure was successfully synthesized.[33,37] The (0 0 2) reflection plane of MC and PMC shifted from 26.4° to a lower value of 21°, implying that the interlayer spacing in the prepared porous carbon materials is expanded from the typical spacing value of graphite (3.34 Å). An intense peak at 18° was observed for MC, ascribed to the stacking structure of polymerized hydroxyl methyl furfuraldehyde and its other aldol condensed carbon rings.[38,39] However, this was not visible in the XRD of PMC, likely due to the phytic acid-catalyzed effective carbonization of the biomasses at high temperature.[33] In terms of morphology, the SEM images revealed agglomerated mesoporous carbon particles (Figure 2c,d). The EDX spectra confirmed the presence of phosphorus in PMC, as reflected by the detected P signal (Figure 2f); this is in contrast to MC (Figure 2e), which only contains peaks for C and O. The textural properties of the prepared materials were further characterized by N_2 adsorption−desorption analysis. The isotherm curves of both materials exhibit a type IV isotherm with an H1 hysteresis loop and sharp capillary condensation at higher relative pressures (Figure 2g), which is typical of mesoporous materials.[33,40] The BET surface area (S_{BET}), pore diameter, and pore volume were calculated, and the results are given in Table S1. The S_{BET} values of MC and PMC were determined to be 1046.5 and 1482.1 m^2 g^{-1}, respectively, along with total pore volumes of 1.46 and 1.70 cm^3 g^{-1}, respectively. This shows that the introduction of phosphorus into the carbon framework results

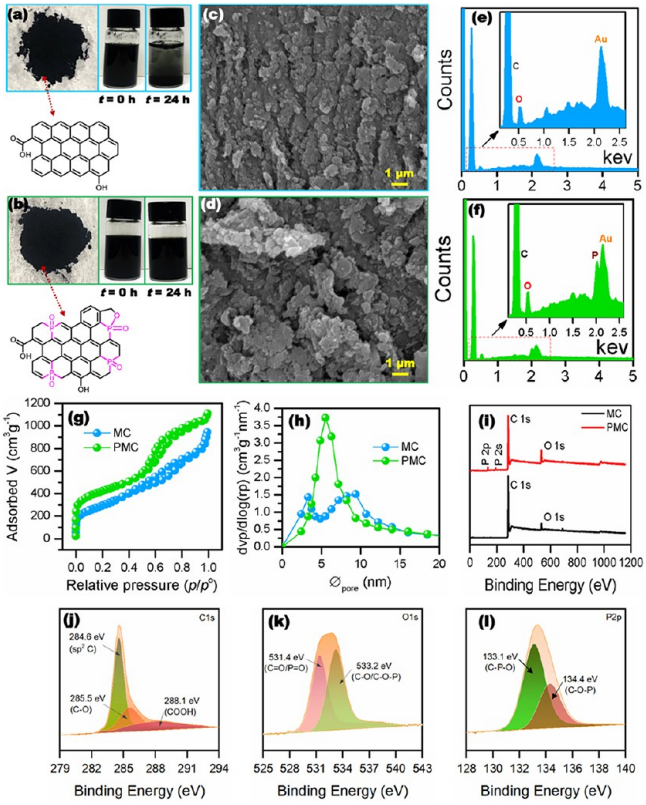


Figure 2. Photographs, molecular structure, and dispersion in water for (a) MC and (b) PMC. SEM−EDX of (c, e) MC and (d, f) PMC. (g) N_2 sorption isotherms and (h) pore size distribution curves of MC and PMC. (i) XPS survey spectra of MC and PMC and high-resolution XPS deconvolution spectra of (j) C 1s, (k) O 1s, and (l) P 2p for PMC.

in an increase in the S_{BET} and pore volume. This could be attributed to the formation of phosphate linkages on the carbon edge and the larger atomic size of P than C, which led to the expansion of the graphitic plane of PMC. A high specific surface area is important for excellent flame retardant properties.[41] Pore size distribution using the BJH method also reveals that the addition of PA during synthesis yields mesoporous carbon materials with uniform pore size (Figure 2h). To study the nature of P bonding on the surface of PMC and to determine the surface chemical composition, XPS analysis was performed. The XPS spectrum of MC only reveals the presence of C and O while that for PMC indicates the presence of P in addition to C and O (Figure 2i). No Si peak was detected in either spectrum, suggesting that KIT-6 was completely removed. The deconvoluted spectra of C 1s, O 1s, and P 2p for PMC are shown in Figure 2j−l, respectively. The C 1s XPS spectra can be divided into three peaks centered at 284.6, 285.5, and 288.1 eV (Figure 2j), assigned to graphitic sp^3 C, C−O, and COOH, respectively.[42] In Figure 2k, the XPS spectra of O 1s can be deconvoluted into two peaks located at 531.2 and 533.2 eV, ascribed to C═O in carbonyl, quinine or COOH and P═O and C−O in phenol, ethers, or C−O−P, respectively (Table S2).[42]

Moreover, the XPS spectra of the P 2p peak (Figure 2l) revealed that the P is covalently bonded with C in the form of C−P−O (133.1 eV), like C_3—P═O or C−O−P (134.3 eV), i.e., CO−P(O)$(OH)_2$, $(CO)_2$−P(O)(OH), and $(CO)(C)_2$−PO (Table S2).[33,42,43] According to Ternero-Hidalgo et al.,[44]

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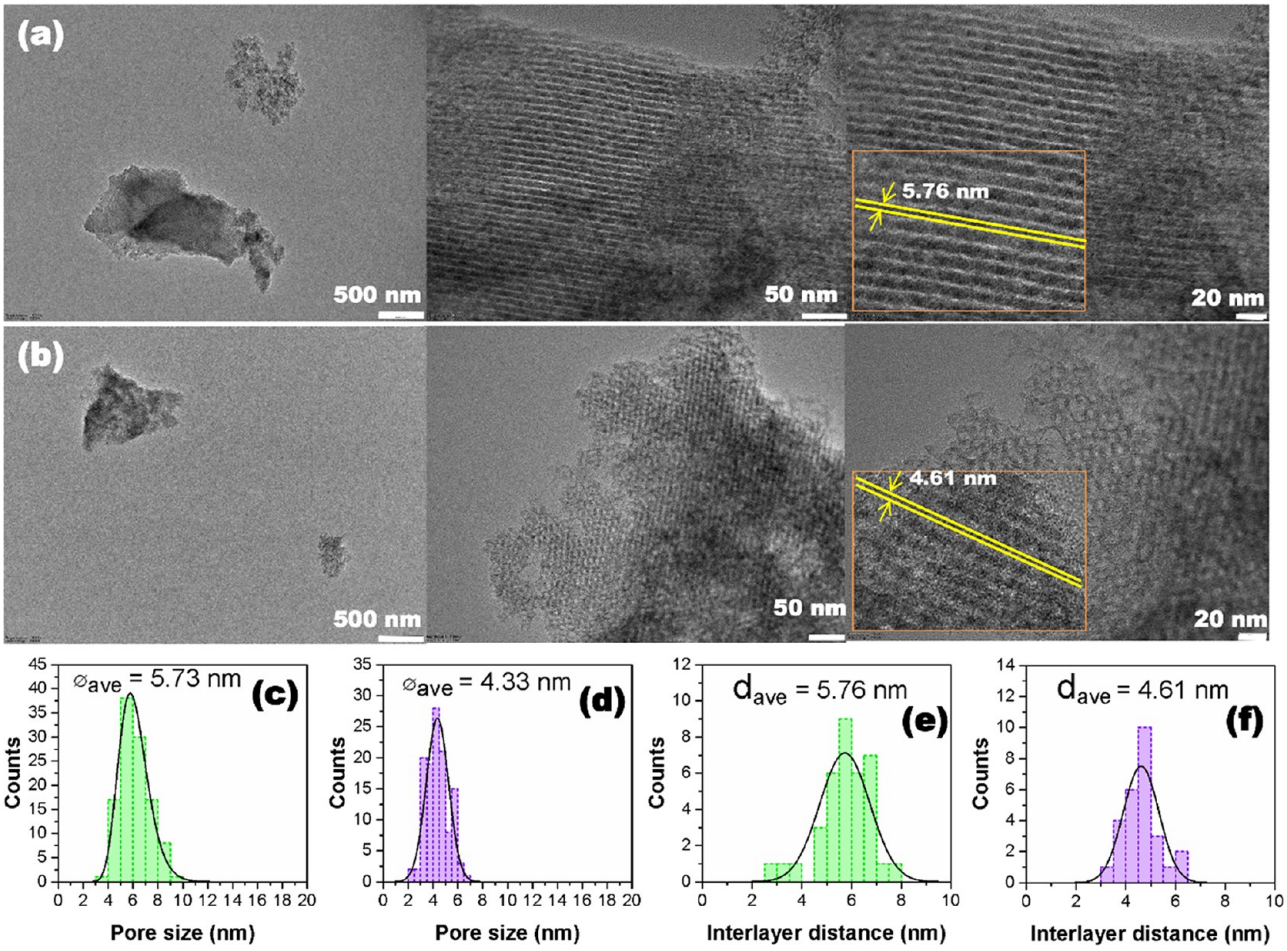


Figure 3. HR-TEM images of (a) MC and (b) PMC. Inset: interlayer distance. Pore size and interlayer distance for (c, e) MC and (d, f) PMC.

the P-containing groups exist in the graphitic structure of PMC, as shown in Figure S3. However, it is not easy to differentiate between these groups via XPS due to their close binding energy. Thus, the bulk elemental (C, H, and O) compositions of both MC and PMC were determined by elemental analysis (EA) and are given in Table S3. A decrease in the C content and an increase in the O content were observed after P doping, confirming the presence of several P-containing oxygenous groups. Furthermore, the materials had a substantial H content, indicating that the materials also contain several sp^2C−H at the edges. After P doping, no increment was observed for the atomic percentage of H, which means that the presence of H containing P-based functional groups (CO−P(O)$(OH)_2$ and $(CO)_2$−P(O)(OH)) in the C−P−O and C−O−P linkages is unlikely. Thus, the P-based moieties are mainly in the form of $(CO)(C)_2$−PO and C_3—P═O. From the deconvoluted P 2p peak, majority of the P is in the form of C_3—P═O. Meanwhile, the decrease in the H content after doping suggests that the P-containing functional groups had partially replaced the sp^2C−H at the edge. Hence, the P is integrated into the edges of graphitic layers through bonds to one or more O atoms. This result was supported by the drastic increase in the P and O contents of PMC (Table S1). Because the P−O linkage occurs specifically on the edges, it also increases the surface area remarkably, as observed during the N_2 adsorption−desorption analysis.

A closer examination of the materials using HR-TEM revealed a long-range ordered mesoporous structure (Figure 3a,b). Compared with that of MC (5.73 nm), the mesopore of PMC shrinks to 4.33 nm (Figure 3c,d), which agrees with the results obtained by the N_2 adsorption−desorption experiment. Additionally, the average interlayer distances were determined to be 5.76 nm for MC and 4.61 nm for PMC (Figure 3e,f).

3.2. Coating Characterization. Digital photographs of the pristine and LbL-coated PUFs are shown in Figure S4. The color darkened as the number of bilayers increased, especially for the PMC-coated PUF. In agreement with the color change, the weight gains were 0.0, 3.41, 8.18, 15.18, and 25.45% for neat PUF, 3 BL MC@PUF, 6 BL MC@PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF, respectively. The higher mass addition for PMC-filled multilayers than MC-filled multilayers for the same deposition layers indicates that the film growth was very effective for PMC particles. This can be explained by the fact that PMC possesses more negative surface charges (see Figure S2) and can be easily dispersed in water in the presence of AL (Figure 2b), which facilitates the self-assembly process. In contrast, MC exhibited poor dispersion in water and began to settle almost immediately after sonication (Figure 2a). Thus, further thin-film characterization experiments were performed with a focus on samples containing PMC multilayers. UV−vis absorption spectroscopy was used to monitor the growth of the CH/AL-PMC multilayer. According to the spectra, the CH and AL-PMC assembled successfully on the substrate, and the deposited amount could be controlled by changing the number of bilayers (Figure 4a). Figure 4b shows that CH and AL-PMC were assembled alternately on the surface. The ATR-FTIR

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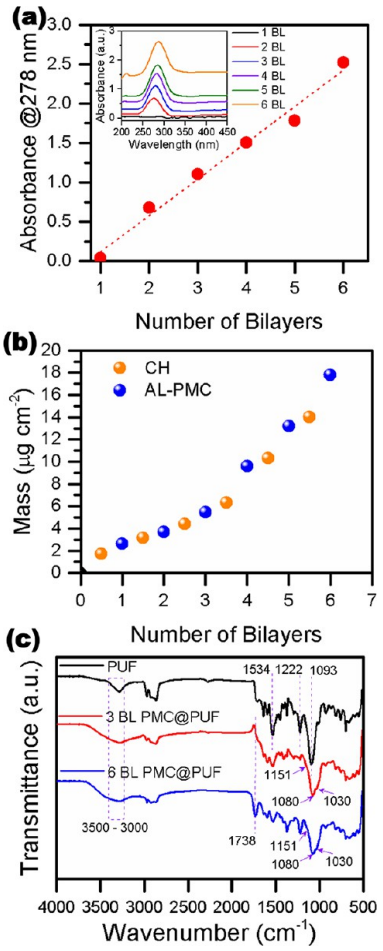


Figure 4. (a) UV−vis and (b) accumulated film mass as a function of deposited bilayers measured by QCM for the CH/PMC-AL film. (c) ATR-FTIR spectra of the neat and PMC coated PUFs.

spectra of the neat and coated PUFs are shown in Figure 4c. The uncoated PUF gave rise to prominent characteristic peaks related to the N−H and C−N vibrations of the urethane and urea groups centered at 1534 cm^{-1} and asymmetric and symmetric C−O−C vibrations at 1222 and 1094 cm^{-1}, respectively.[45] Meanwhile, the successful assembly of the FR materials on PUF was confirmed by the emergence of additional peaks at 1030 and 1738 cm^{-1}, respectively, ascribed to the stretching vibrations of the C−O−C in CH and AL and C═O groups in AL,[22] in addition to the 1080 and 1151 cm^{-1} peaks, which were assigned to the absorbance of PO_3^{2-} and P═O in PMC,[16] respectively. Compared to 3 BL PMC@PUF, the IR peak for C═O groups (1738 cm^{-1}) in AL was more intense for 6 BL PMC@PUF due to the increased concentration of AL at higher deposition layers. Furthermore, the IR signal at 3500−3000 cm^{-1}, which was assigned to the valence vibrations of the N−H and O−H bonds in the amino group of CH and hydroxyl group of AL, respectively, intensified and broadened with increasing bilayers. Thus, the ATR-FTIR results confirm the successful deposition of FR materials on PUF. To observe the morphology changes and structure of the foams, SEM analysis was performed for neat PUF, 3 BL MC@PUF, and 3 BL PMC@PUF. Low-magnification SEM images revealed that both coating systems maintain the microporous open-cellular structure of the foam. At high magnification, neat PUF displayed a clean and smooth surface (Figure 5a−c). 3 BL MC@PUF and 3 BL PMC@PUF had highly rough surfaces, mainly due to the presence of mesoporous carbon particles (Figure 5d−l). The 3 BL MC@PUF contained some bare regions due to the poor dispersion of MC in water. In contrast, the surface of 3 BL PMC@PUF appeared to be entirely covered by PMC particles embedded within the CH and AL matrices (Figure 5k,l), with some aggregated PMC particles on top of this coating (Figure 5i,j).

3.3. Thermal Stability. The thermo-oxidative behavior of neat and coated PUFs was investigated by TGA in air. The TGA and DTG curves are shown in Figure S5a,b, and the relevant thermal parameters are summarized in Table S4. The thermal degradation of the uncoated PUF occurs in two stages: the first stage at around 295 °C is caused by the pyrolysis of

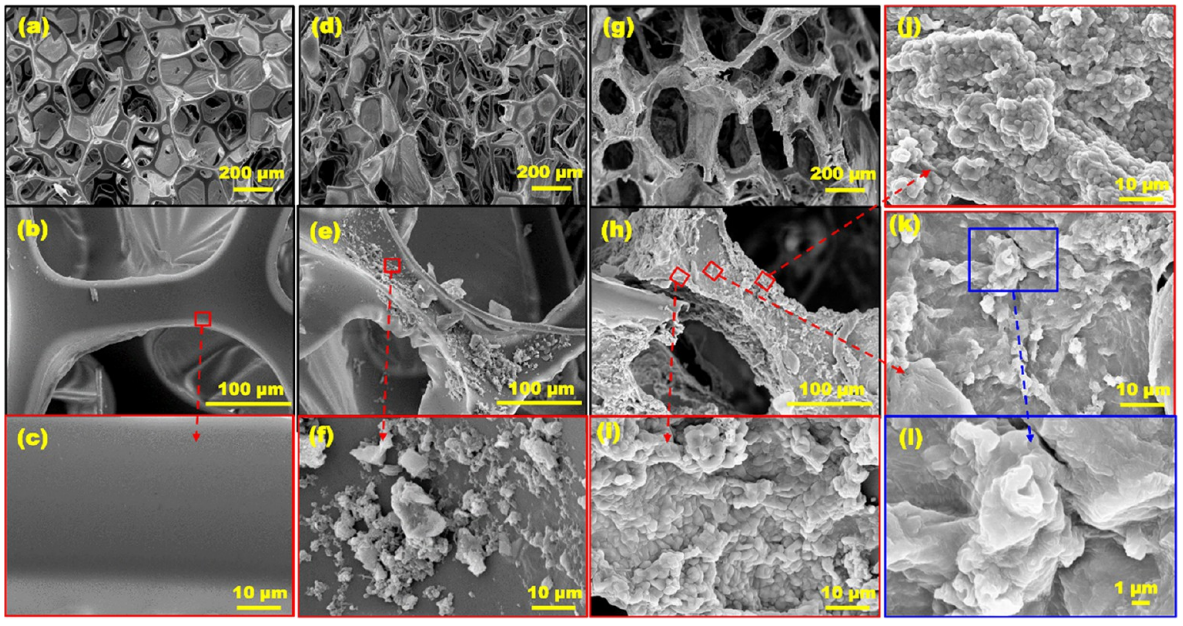


Figure 5. (a−c) SEM images of uncoated PUF, (d−f) 3 BL MC@PUF, and (g−l) 3 BL PMC@PUF.

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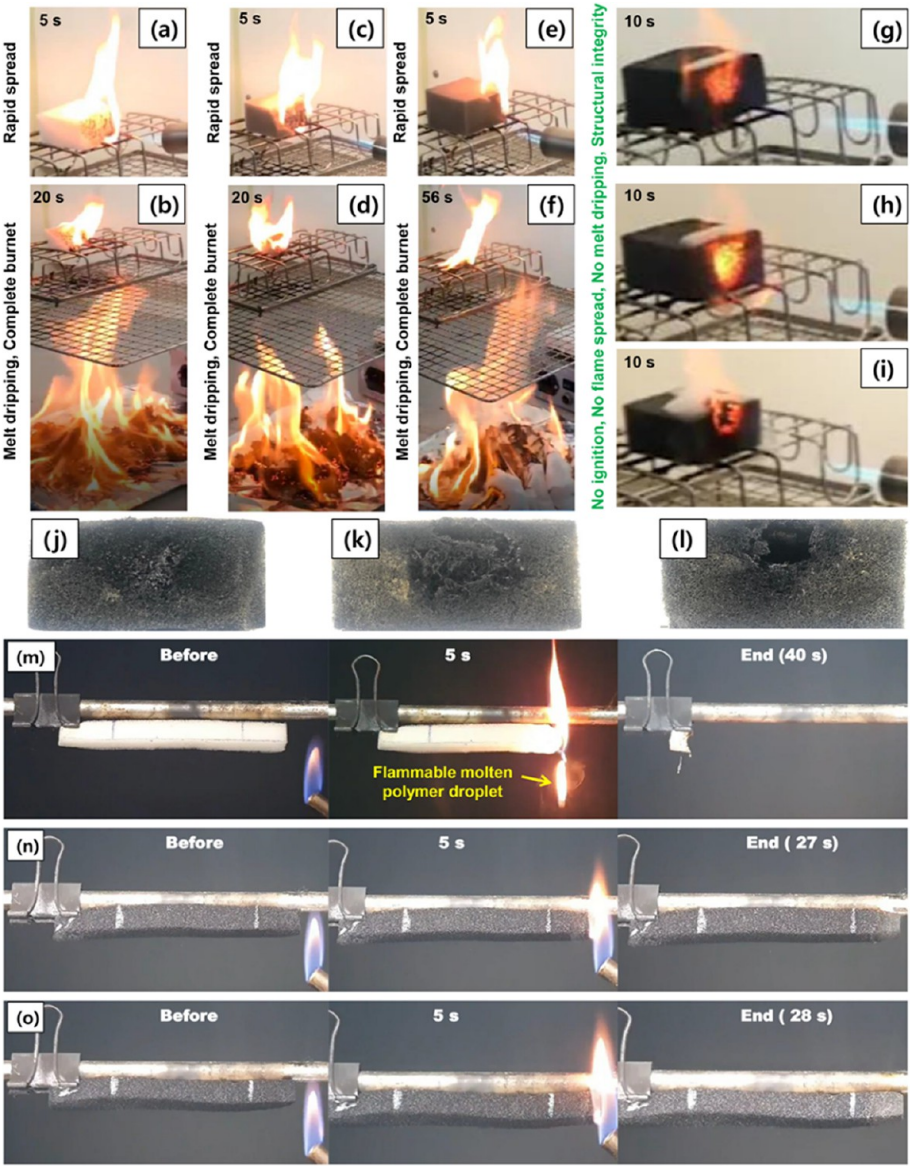


Figure 6. Digital images of neat and coated PUFs after different times during the open flame test. (a, b) PUF, (c, d) 3 BL MC@PUF, (e, f) 6 BL MC@PUF, (g) 3 BL PMC@PUF, (h) 6 BL PMC@PUF, and (i) 3 BL PMC@PUF after 500 compression cycles. The cross sections after flameout: (j) 3 BL PMC@PUF, (k) 6 BL PMC@PUF, and (l) 3 BL PMC@PUF after 500 compression cycles. Digital images during the UL-94 horizontal burning test. (m) Neat PUF, (n) 3 BL PMC@PUF, and (o) 6 BL PMC@PUF.

urethane and urea hard segments, while the other stage at 328 °C arises from the decomposition of the polyether soft segment.[12] After PMC-filled coatings, the T_{max2} completely disappeared, indicating that a stable char layer that can effectively hamper further decomposition was formed during the first decomposition step. Compared to the pure PUF, the 5 wt % loss temperature ($T_{-5\%}$) of the 3 BL PMC@PUF and 6 BL PMC@PUF decreased due to the early decomposition of the coatings. However, the decomposition rate in the high temperature range (350−800 °C) of the coated PUF is decreased significantly, as evidenced in the DTG curve (Figure S5b). Other previously reported phosphorous-based or expandable graphite FRs also often led to a decrease in the $T_{-5\%}$ value of the foam.[12] Furthermore, the char residues remained at 700 °C for these samples were significantly higher than those of the uncoated PUF, indicating that the coatings enhanced the thermal stability and char formation at high temperatures.

3.4. Flame Retardant Properties. A flammability test was performed to screen the FR behavior of the coated PUFs prior to further fire testing and determine their contribution in initiating a fire. Videos showing the burning behavior of each PUF sample are given in the Supporting Information, while Figure 6a−l shows the digital photographs of the samples during the torch test. Upon exposure to a butane flame, the neat PUF immediately ignited, generated a dropping, flaming melt pool, and was completely consumed without any remaining residue while the fire was transferred to the flammable material beneath (Video S1 and Figure 6a,b). This phenomenon is characteristic of flexible foam and is a major concern because it tends to transfer fire to neighboring flammable materials, that is, textiles and carpet in a real house-fire scenario.[1,45] Similar results were observed for MC-filled coatings regardless of the number of deposited BLs, except that the melt-dripping time was prolonged for the 6 BL-coated PUF (Videos S2 and S3 and Figure 6c−f). The low FR performance

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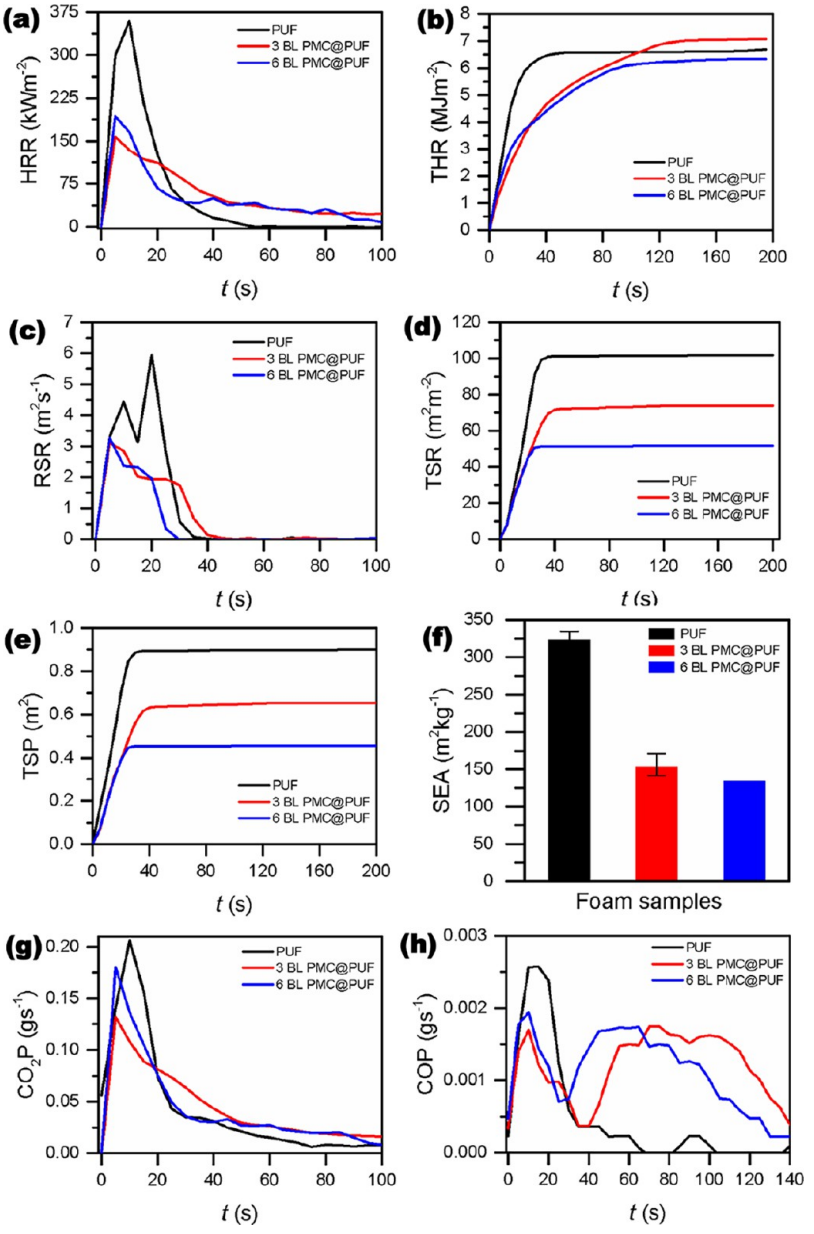


Figure 7. (a) HRR, (b) THR, (c) RSR, (d) TSR, (e) TSP, (f) SEA, (g) CO_2P, and (h) COP as a function of time for pristine (control) and coated PUFs.

of MC-filled coatings could be attributed to the poor deposition of MC on the PUF surface (as confirmed by the SEM images in Figure 5f). In contrast, the presence of PMC-filled coatings (3 BL PMC@PUF and 6 BL PMC@PUF) radically changed this behavior. Indeed, these coatings completely prevented flame flashover, melt-dripping, and fire propagation to nearby materials by immediately extinguishing the fire after the flame was removed (Videos S4 and S5 and Figure 6g,h). These results clearly show that the introduction of phosphorus into the porous carbon framework improves the dispersion in water and thin-film formation of PMC onto the surface of PUF, thereby enhancing the flame retardancy. As shown in the cross-sectional images (Figure 6j,k), these foams were only charred where the flame contacted it, losing only ≈1.1−2.2 wt % of their initial mass. Even after 500 cycles of compression (at 60% strain), the foam coated with 3 BL PMC retained its original size and shape (Figure S5c−e) and its ability to self-extinguish (Video S6 and Figure 6i). This indicates that the coating was stable and had a minor effect on the mechanical properties of the foam. It is worth noting that the foam after compression was incapable of withstanding flame penetration (Figure 6l). The morphology of the char layer after the open flame test before and after the compression test was observed by SEM (Figure S6). High-magnification SEM images revealed an intumescent-like, rough morphology. Thus, the PMC-filled LbL coating improved the fire safety of PUF owing to the formation of a rough, protective char layer that prevented foam shrinkage, flame spread, melt-dripping, and fire propagation to a neighboring flammable material.

The open flame test using a hand-held torch serves as a proof of concept and is helpful for screening FR foams before sending for further fire standard tests. However, its reproducibility might be limited due to the lack of proper control of the burning conditions, i.e., fire load, the distance

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between torch and foam samples, etc. Therefore, the flammability of pristine PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF was further evaluated using the standard UL-94 horizontal burning test. Obviously, PUF burned violently, generated a flammable molten polymer capable of transferring fire to the material beneath, and was utterly burned (Figure 6m and Video S7). Hence, neat PUF fails the HB rating, and it imposes a very high fire threat. In contrast, the UL-94 rank of foams modified with PMC-filled coatings (3 BL PMC@PUF and 6 BL PMC@PUF) reaches HB as these foams avoid violent burning and melt-dripping and self-extinguish immediately when the flame is removed (Figure 6n,o, and Video S7), which is inconsistent with the results observed in the open flame test. These foams were only charred where the flame touched, and they have maintained ∼97% of their initial weight after the burning process (Table S5). Such excellent FR property is favorable for the practical application of flexible PUFs.

Cone calorimetry testing provides a complete set of data related to the contribution of the material to fire growth and smoke production. Therefore, the neat and coated foams were characterized by applying a heat flux of 35 kW m^{-2}, which resembles the initial phases of a developing fire. Since MC-based coatings were ineffective, the CC test was only carried out for neat PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF. The neat PUF ignited quickly (TTI ∼2 s), yielding a sharp pkHRR of 359 KW m^{-2} (Figure 7a and Table S6). In contrast, it took slightly longer to ignite the PMC-filled coatings and yielded a significant reduction in the pkHRR compared to the neat foam. In particular, the 3 BL PMC-filled coating yielded the lowest pkHRR (158 kW m^{-2}) among all samples, which is 56% lower than that of neat PUF. However, the reduction in pkHRR did not increase after the bilayer number was increased to six (i.e., 6 BL PMC@PUF). This indicates that the 3 BL CH/AL-PMC coating was sufficient enough to play an effective role in improving the flame retardancy of PUF, and more bilayers would not contribute to the flame retardancy further. This phenomenon could be explained as follows: The LbL assembly initially grows linearly and then exponentially after the third BL (see QCM results Figure 4b). Given that the deposition of one PMC layer needs one layer of CH and one layer of AL, the concentration of CH and AL might be higher at an increased deposition layer, which leads to a higher pkHRR. The pkHRR of 3 BL and 6 BL CH/AL alone (without PMC) coated PUF was determined. The CH/AL alone coatings exhibited slightly higher pkHRR values than that of neat PUF (Figure S7), which is believed to be due to the flammability of the biopolymers.[1,3] Hence, the contribution of CH and AL to the flame retardancy of PUF is insufficient, except for acting as effective film-forming agents in the binding of PMC to the surface of the PUF. Usually, non-intumescent polymer-based FR coatings are not effective for PUF unless they are combined with an appropriate amount of particle-based, i.e., clay or carbon FRs.[1,46] Other LbL-deposited FRs on PUF, such as CH/GO and CH/ammonium polyphosphate, have also led to a decrease in pkHRR reduction upon increasing the number of bilayers.[4,22] Overall, the results showed that the PMC-filled coating significantly reduced pkHRR, one of the most important fire safety metrics that represents the point in a fire where the flame likely propagates to adjacent flammable materials. A lower pkHRR value implies a lower risk of spreading to neighboring flammable items, thereby providing more time for evacuation from a real fire incident.[47] The fire growth index rates (FIGRA = pkHRR/t_{pkHRR}) for all foam samples are given in Table S6. Generally, a lower value of FIGRA indicates better fire-safety performance.[48] As indicated in Table S6, 3 BL PMC@PUF achieved the lowest FIGRA value (31.6 kW m^{-2} s^{-1}), which is 12% lower than that of uncoated PUF, further confirming that this coating significantly improved the fire safety of PUF. This coating also gave the lowest maximum average rate of heat emission (MARHE) among all samples. The total heat release (THR) curves of the pristine and coated PUFs are shown in Figure 7b. Only 40 s is required for the THR curve of neat PUF to reach the plateau because of its high flammability. On the other hand, approximately 95−125 s is needed to reach the plateau for PMC-filled coatings, implying that the presence of PMC delayed the THR.

The THR values of pristine PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF were 6.7, 7.1, and 6.4 MJ m^{-2}, respectively. The standard deviation of these three values is 0.35, meaning that the coatings have no significant effect on the THR, likely due to the additional heat release of the coating materials.[49] Other reported CH/lignosulfonate,[49] hexagonal boron nitride/alginate/polyethylenimine,[50] CH/GO,[22] aluminum hydroxide/polyethylenimine,[3] CH/ammonium polyphosphate or CH/montmorillonite,[4] and silicone/reduced GO[51] coatings also showed a similar phenomenon.

When PUF is ignited, it produces an enormous amount of harmful smoke and gases, which are the main causes of fire-related deaths.[52,53] Therefore, the smoke and gas production-related parameters, including the rate of smoke release (RSR), total smoke release (TSR), total smoke production (TSP), smoke optical density (SEA), carbon dioxide production rate (CO_2P), and carbon monoxide production rate (COP), are considered as additional key metrics to evaluate the PUF FR performance. According to Figure 7c and Table S6, the peak RSR (pkRSR) values for neat PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF were 5.9, 3.1, and 3.3 m^2 s^{-1}, respectively, with reductions of 47.5 and 44.1% compared to neat foam. In addition, the PMC-filled coatings exhibit a substantial reduction in TSR and TSP. Compared to neat PUF, the TSR and TSP values of the 3 BL PMC and 6 BL PMC samples were ≈29 and 49% lower, respectively (Figure 7d,e and Table S6). Furthermore, the 3 BL and 6 BL PMC-coated samples considerably reduced the SEA by 52 and 58%, respectively (Figure 7f). This remarkable reduction indicates that the PMC-filled coating not only effectively reduces the combustion rate and fire spread hazard but also offers an excellent smoke suppression effect. Indeed, the release of less-dense smoke simplifies the evacuation and firefighting processes during a fire incident. The CO_2P and COP of the foam samples are shown in Figure 7g and Figure 7h, respectively. The peak CO_2P ($pkCO_2P$) values for the 3 BL and 6 BL PMC-coated samples were 0.13 and 0.18 g s^{-1}, respectively, with diminishments of 35 and 10%, respectively, further confirming the excellent FR properties of the PMC-filled coatings. This outstanding property was also observed in the suppression of peak COP (pkCOP) with reductions of 35 and 27% for 3 BL and 6 BL PMC-coated PUF, respectively. The average effective heat combustion (av-EHC) value is a useful parameter for analyzing the FR mechanism in the gas phase. A low av-EHC value indicates the presence of non-combustible gases in the gas phase, thereby supporting a gas-phase FR mechanism through fuel dilution.[54−56] The av-EHC of 3 BL PMC@PUF is 18.4 MJ kg^{-1}, which is similar to that of the pristine PUF (18.7 MJ

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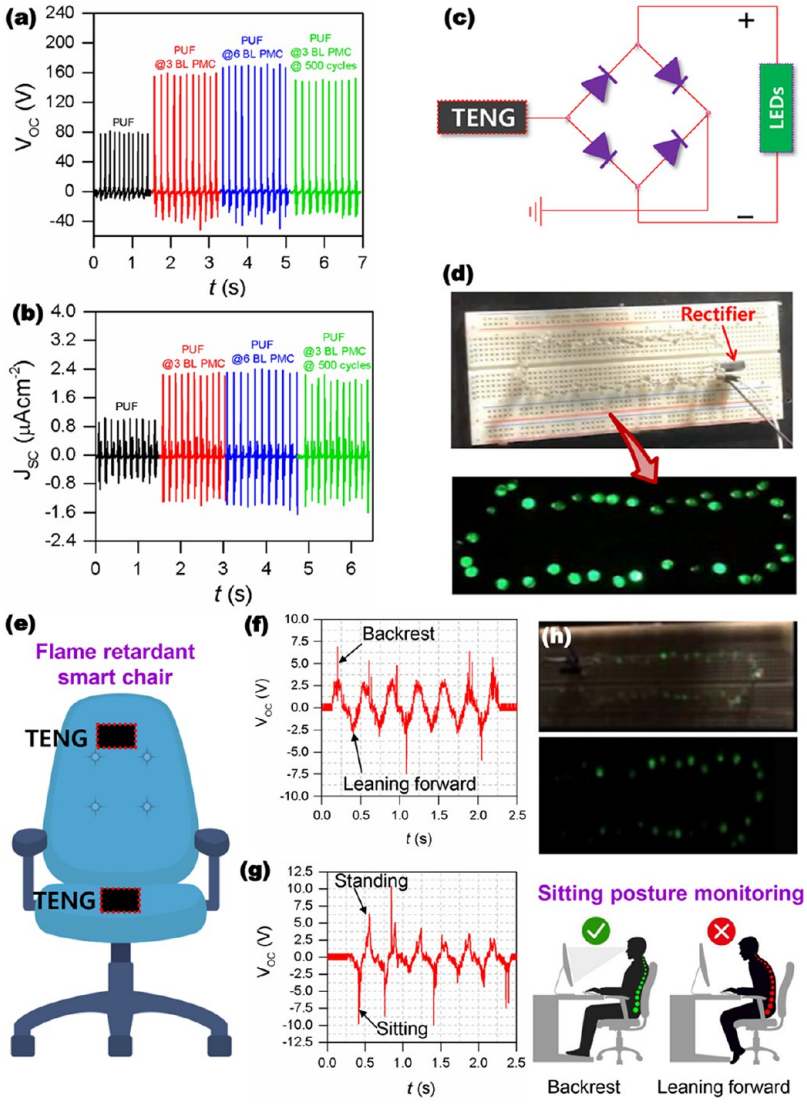


Figure 8. (a) V_{OC} and (b) J_{SC} outputs for neat and coated PUF. (c) Circuit diagram for powering commercial LEDs. (d) Lighting up 38 LEDs with manual hand tapping. Human motion detection applications of the FRF-TENG. (e) FRF-TENG adhered to the back rest and seat plate of a chair. V_{OC} output to the motions of (f) back movement and (g) sit-stand motion. (h) Twenty-one LEDs powered by back movement.

kg^{-1}). This suggests that the improved FR performance of 3 BL PMC@PUF is mainly due to the formation of a protective char layer that blocks the exchange of materials (oxygen, combustion gases, etc.) and heat between the condensed and gas phases.

The digital images of the foam leftover after CC analysis are depicted in Figure S8a−c. The surfaces of the char residues of 3 BL PMC@PUF and 6 BL PMC@PUF foams were brown with a fluffy black char inside. After the CC test, only 2.8% of its initial weight was left for the uncoated PUF (Figure S8d and Table S6) due to its poor FR performance. In contrast, the char residues that remained after the CC combustion for 3 BL PMC@PUF and 6 BL PMC@PUF were 49.8 and 46.9 wt %, respectively (Figure S8d and Table S6). These results suggest that the PMC-filled coatings favor char residue formation, thereby preventing materials and heat transfer between the condensed and flame zones. A high char yield means lesser fuel in the flame zone, and the FR additives mainly function in the condensed phase to produce a protective char layer.[57]

3.5. Application as the Triboelectric Nanogenerator. Foams are excellent choices for the fabrication of energy-harvesting devices and self-powered sensors aside from their application in seat cushions, mattresses, packaging, vehicle seats, heat, electricity, and noise insulators. Thus, using the PMC-coated foam as dielectrics, the flame retardant foam TENG (FRF-TENG) with single-electrode mode can be fabricated. The layout of a typical single-electrode mode FRF-TENG setup with 3 BL PMC@PUF as the positive tribolayer and PTFE as the negative tribolayer is shown in Figure S9. Details on the electricity generation principle of the FRF-TENG can also be found in the Supporting Information p. S16-17. Control experiments were also designed using the neat and 6 BL PMC-coated foams to test the effect of surface modification and number of deposited layers on the triboelectric performance of PUF. The open-circuit voltage (V_{OC}) and short-circuit current density (J_{SC}) of the neat and PMC-coated PUFs are shown in Figure 8a and Figure 8b, respectively. Neat PUF exhibited a V_{OC} of 80 V and a J_{SC} of 1.0 μA cm^{-2}. The V_{OC} and J_{SC} increased to 158 V and 2.26 μA cm^{-2}, respectively, for 3 BL PMC@PUF and 170 V and 2.38 μA cm^{-2}, respectively, for 6 BL PMC@PUF FRF-TENGs. This significant output performance improvement can be

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attributed to the high surface roughness and high tribo-positive charges from the nitrogen, oxygen, and phosphorus abundant coating materials, i.e., CH, AL, and PMC. It is worth noting that the 3 BL PMC@PUF has excellent flame retardancy, comparable TENG performance, and minimum number of layers compared to the 6 BL PMC@PUF. Thus, 3 BL PMC@PUF was used in the subsequent experiments. To display the electric output visually, the FRF-TENG was connected to light-emitting diode (LED) bulbs (3.4 V forward voltage and 100° viewing angle at 50% IV) through a bridge rectifier, as depicted in Figure 8c. The FRF-TENG could lit up 38 green LEDs (Figure 8d and Video S8). Mechanical durability of the FR-TENG is essential as the TENG is subjected to considerable mechanical stress in practical applications. Hence, we investigated the durability of the FRF-TENG first by conditioning the foam (compression of the 3 BL PMC@PUF for 500 times) and then measuring the output voltage and current density. Compared to the uncompressed 3 BL PMC@PUF, no significant degradation was observed in the output performance after the 500 cycles of deformation (Figure 8a,b), confirming its excellent durability. The practicability of the FRF-TENG for human motion sensing applications was also highlighted. Given that PUF can be easily integrated into seat and transport chairs, an FRF-TENG (with 5 × 5 cm^2 effective contact area) was attached with adhesive tape on the back rest and seat plate (Figure 8e) of a chair to detect the movement of back and sit-stand motion, respectively. The highest voltages of up to ≈7 V (Figure 8f) and ≈10 V (Figure 8g) were achieved during cyclic leaning forward−backward of the back and sit-stand movements, respectively. Furthermore, the power generated during back movement can light up more than 20 green LEDs, as shown in Figure 8h and Video S9. This means that the FRF-TENG can efficiently power small electronic devices when the back moves, and the lit LEDs can alert people to correct their sitting posture. Such delicate sensing abilities of a foam can ultimately enable the fabrication of a smart chair that can monitor sitting posture, child or disabled people safety, and traffic safety. The performance of the FRF-TENG after fire attack was explored by exposing the FRF-TENG to a torch for 10 s. The FRF-TENG immediately self-extinguishes after removing the flame with only charring at the area where the fire contacted. Furthermore, the FRF-TENG successfully powered the LEDs without reducing luminance even after the flame attack (Figure S10 and Video S10). This result implies that the FRF-TENG could monitor human motions even in a fire hazard, e.g., movement of firefighters during a fire rescue.

4. CONCLUSIONS

A PMC-filled multilayer was successfully deposited on PUF using an LbL technique. This involved the preparation of PMC from bio-sourced saccharose and phytic acid using mesoporous silica KIT-6 as a hard template via the hydrothermal method. The coated PUF was then prepared by alternately submerging PUF into a positively charged chitosan solution and negatively charged alginate-stabilized PMC aqueous dispersion. ATR-FTIR, UV−vis, QCM, and SEM indicated that the coating was effectively built on the surface of the foam, while TGA revealed the effectiveness of the coatings in enhancing the thermo-oxidative stability of PUF. A few layers (three bilayers) of the CH/AL-PMC coating enabled the foam to self-extinguish when exposed to a butane flame (≈1400 °C). Additionally, this coating layer warranted the foam to pass the UL-94 rating. Cone calorimetry revealed that this coating reduced the pkHRR, RSR, TSR, peak CO_2 production rate, and peak CO production rate by 56, 48, 29, 35, and 35%, respectively. Using this foam, a flame retardant foam-based triboelectric nanogenerator (FRF-TENG) was fabricated. The FRF-TENG exhibited excellent energy harvesting performance, generating an open-circuit voltage of 158 V and short-circuit current density of 2.26 $\mu A\ cm^{-2}$. The TENG could be attached on a chair, generating outputs of 7 and 10 V upon back movement and sit-stand motion, respectively. The output was sufficient to concomitantly glow 21 LEDs, demonstrating that the FRF-TENG can be used as a self-powered sensor for human motion detection. This study demonstrates a straightforward, low-cost, and fully renewable nanocomposite coating for the design of multifunctional smart foams.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsanm.2c01999.

XRD and Zeta potential of MC and PMC, N_2 adsorption−desorption and XPS results, elemental analysis, digital images of neat and coated PUF, TGA and mechanical properties, SEM image of the char residue, summary of UL-94 horizontal burning and cone calorimetry results, TENG layout, and electricity generation mechanism (PDF)

Torch test video of neat PUF (AVI)

Torch test video of 3 BL MC@PUF (AVI)

Torch test video of 6 BL MC@PUF (AVI)

Torch test video of 3 BL PMC@PUF (AVI)

Torch test video of 6 BL PMC@PUF (AVI)

Torch test video of 3 BL PMC@PUF after 500 cycle compression (AVI)

UL-94 horizontal burning test for neat PUF, 3 BL PMC@PUF, and 6 BL PMC@PUF (play speed: 5×) (AVI)

LEDs lighting up by the FRF-TENG (play speed: 2×) (MP4)

Demonstration of the back movement detection (play speed: 2×) (MP4)

Powering LEDs after fire attack (play speed: 1.5×) (AVI)

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Author Contributions

All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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